Credit Name: CSE 3130 Object-Oriented Programming 2.
Assignment Name: Vehicle

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem:
How did you approach understanding the challenge?

- So I understand that in this project I need to use an abstract method to come up with the information for 3 different vehicles.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

- This was my first project using abstract, so it took me a while to figure out how I can apply the abstract in my project. Basically like how an abstract method will look like and what is the difference between abstract method and normal method?
- I solved this problem by looking for the example in the textbook and also some more examples in OverflowStack about the structure and the difference.

Planning the Solution:
Did you create any or break the problem into smaller steps before coding?

- I tend to make it easy by just doing it step by step and checking it before I move on so that I can keep track of the bugs and fix it instantly if there are any.

How did you decide on the tools, data structures, or algorithms to use?

- I first created the Vehicle class as the prompt asks for. With the abstract method in the getInformation() which I will update in each specific class.
- Then I worked in each class included: Car, Minivan, and Truck as assigned.

Implementation:
Did you write the code in small or attempt the entire solution at first?

- I broke the project into smaller like this:
    - Vehicle:
        - ```
          public Vehicle (double city, double hwy, int capacity, double volume)
              fuelEconomyCity = city; //Sets the city fuel economy//
              fuelEconomyHwy = hwy; //Sets the highway fuel economy//
              seatingCapacity = capacity; //Sets the seating capacity//
              cargoVolume = volume; //Sets the cargo volume//
          }
          ```
        - I first assigned what values will be included when the data of the vehicle is entered.
        - ```
          public abstract String getInformation(); /
          ```
        - Then I created an abstract method to return the information of the specific vehicle.
    - Car/Truck/Minivan

- ```java
  public Car(double city, double hwy, int capacity, double volume) {
      super(city, hwy, capacity, volume); //Calls Vehicle's construct
  }
  ```
- First I called all the values in the parent class and extended each of these classes so that I can also call for the values and methods in the parent class.
- ```java
  public String getInformation() {
      String information = "-----------------------------------\n" +
                      "•Vehicle: Car.\n" +
                      "•Economy City Fuel: " + fuelEconomyCity + " L/100Km.\n" +
                      "•Economy Highway Fuel: " + fuelEconomyHwy + " L/100Km.\n"
                      "•Capacity: " + seatingCapacity + " seats.\n" + //Displays
                      "•Cargo Volume: " + cargoVolume + " CBM.\n" + //Displays c
                      "-----------------------------------\n";
      return information;
  }
  ```
- Then I updated the getInformation() method to return the information of the vehicle.

○ Client_test

- ```java
  Vehicle car = new Car(28, 39, 5, 15.1); //Assume values for car//
  Vehicle truck = new Truck(20, 26, 3, 62.3); //Assume values for truck//
  Vehicle minivan = new Minivan(19, 28, 8, 32.8); //Assume values for minivan//
  ```
- First I assigned each vehicle with specific numbers about their Fuel Volume, capacity and cargo volume.
- ```java
  System.out.println(car.getInformation()); //Pr
  System.out.println(truck.getInformation());
  System.out.println(minivan.getInformation());
  ```
- Finally I printed out the information of each vehicle.

How did you test your solution along the way to make sure it was working?

- This was a very straightforward project, so I basically did the entire first and then I tested the entire thing out after.
- I tested the project after I finished the Car class and tested if the data I put in could be shown or the data is correct or not.

Overcoming Challenges:

What part of the problem was the most difficult for you?

- I did not know how to work with abstract methods, so it actually took me more time to figure it out than actually doing the entire project. Because the format is kind of different and new for me to work with

Learning:

Was there anything you learned that you think will help you with future challenges?

- I think abstract is a good way to flexibly adjust the information within the sub class without depending on the algorithm from the parent class. By this, I think it is better to create algorithms that are different from each sub class.